

Binomial Expansion

$$(a + b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \binom{n}{3}a^{n-3}b^3 + \dots$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Maclaurin Expansion

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^n}{n!}f^{(n)}(0) + \dots$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)\cdots(n-r+1)}{r!}x^r + \dots = \sum_{r=0}^{\infty} \binom{n}{r}x^r \quad (|x| < 1)$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^r}{r!} + \dots = \sum_{r=0}^{\infty} \frac{x^r}{r!} \quad (\text{all } x)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^r x^{2r+1}}{(2r+1)!} + \dots = \sum_{r=0}^{\infty} \frac{(-1)^r x^{2r+1}}{(2r+1)!} \quad (\text{all } x)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^r x^{2r}}{(2r)!} + \dots = \sum_{r=0}^{\infty} \frac{(-1)^r x^{2r}}{(2r)!} \quad (\text{all } x)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + \frac{(-1)^{r-1} x^r}{r} + \dots = \sum_{r=1}^{\infty} \frac{(-1)^{r-1} x^r}{r} \quad (-1 < x \leq 1)$$

Partial Fraction Decomposition

$$\frac{px+q}{(ax+b)(cx+d)} = \frac{A}{ax+b} + \frac{B}{cx+d}$$

$$\frac{px^2+qx+r}{(ax+b)^2(cx+d)} = \frac{A}{ax+b} + \frac{B}{(ax+b)^2} + \frac{C}{cx+d}$$

$$\frac{px^2+qx+r}{(ax+b)(x^2+c)} = \frac{A}{ax+b} + \frac{Bx+C}{x^2+c}$$

Trigonometry

$$\sin(A \pm B) \equiv \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) \equiv \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) \equiv \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A \equiv 2 \sin A \cos A$$

$$\cos 2A \equiv \cos^2 A - \sin^2 A \equiv 2 \cos^2 A - 1 \equiv 1 - 2 \sin^2 A$$

$$\tan 2A \equiv \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin P + \sin Q \equiv 2 \sin\left(\frac{P+Q}{2}\right) \cos\left(\frac{P-Q}{2}\right)$$

$$\sin P - \sin Q \equiv 2 \cos\left(\frac{P+Q}{2}\right) \sin\left(\frac{P-Q}{2}\right)$$

$$\cos P + \cos Q \equiv 2 \cos\left(\frac{P+Q}{2}\right) \cos\left(\frac{P-Q}{2}\right)$$

$$\cos P - \cos Q \equiv -2 \sin\left(\frac{P+Q}{2}\right) \sin\left(\frac{P-Q}{2}\right)$$

Principal values:

$$-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2} \quad (|x| \leq 1)$$

$$0 \leq \cos^{-1} x \leq \pi \quad (|x| \leq 1)$$

$$-\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2}$$

Vectors

The point dividing AB in the ratio $\lambda : \mu$ has position vector $\frac{\lambda b + \mu a}{\lambda + \mu}$

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

2d calculus

Absolute value

- $|xy| = |x||y| \quad \forall x, y \in \mathfrak{R}$
- $\sqrt{x^2} = |x| \Leftrightarrow |x|^2 = x^2 \quad \forall x \in \mathfrak{R}$
- $|x+y| \leq |x| + |y| \quad \forall x \in \mathfrak{R}$ (Triangle Inequality)

Functions

- Injective:** For any $x, y \in A$, $f(x) = f(y) \Rightarrow x = y$. (One-one)

- Surjective:** For any $z \in B$, $\exists x \in A$ where $f(x) = z$. (Range = Co-Domain)

- Maximal domain/range:** The largest domain/range that the function can have in $\mathfrak{R}$.

- Squeeze theorem:** Suppose I is an interval that contains point c and $g(x) \leq f(x) \leq h(x)$ for x near c .

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L \Rightarrow \lim_{x \rightarrow c} f(x) = L$$

- Intermediate Value theorem:** Suppose f is continuous on $[a, b]$ and $f(a) \leq k \leq f(b)$. This implies that $\exists c \in [a, b]$, $f(c) = k$.

Limits

- $\lim_{x \rightarrow c} f(x)^{g(x)} = e^{\lim_{x \rightarrow c} g(x) \ln f(x)}$

If $\lim_{x \rightarrow c} g(x) = 0$:

$$\lim_{x \rightarrow c} \left(\frac{\sin g(x)}{g(x)} \right) = 1$$

$$\lim_{x \rightarrow c} \left(\frac{\tan g(x)}{g(x)} \right) = 1$$

Derivatives

- Definition:**

$$\begin{aligned} f'(x_0) &= \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \\ &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \text{by letting } x = x_0 + h \end{aligned}$$

- Inverse function:**

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

- Concavity:**
 - If $f''(c) > 0$, f is concave up at c .
 - If $f''(c) < 0$, f is concave down at c .

- Critical point:** Both must be satisfied.

- c is **not** an endpoint
- $f'(c) = 0$ or $\nexists f'(c)$

- Extreme value theorem:** If f is continuous on $[a, b]$, then $\exists$ a local maximum and minimum in the range $[a, b]$. Note that this will not hold if interval is not of the form $[a, b]$ (e.g. $(a, b]$)

- L'Hopital's rule:** If $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} f(x) = 0 \vee \pm \infty$,

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

- Rolle's theorem:** Let f be differentiable on $[a, b]$, where $f(a) = f(b)$. Then there $\exists c \in [a, b]$ where $f'(c) = 0$.

- Mean Value Theorem:** Let f be differentiable on $[a, b]$. There $\exists c \in [a, b]$ where $f'(c) = \frac{f(b)-f(a)}{b-a}$. This is because the chord between a and b is parallel to the tangent at c .

Integrals

- FTC 1:** $\int_b^a f(x)dx = F(a) - F(b)$

- FTC 2:** $\frac{d}{dx} \int_a^x f(t)dt = f(x)$; more generally, $\frac{d}{dx} \int_a^{g(x)} f(t)dt = f(g(x))g'(x)$

- Area between 2 curves:** $A = \int_b^a |f(x) - g(x)| \, dx$

- Volume for rotation:**

- Shell method:

$$V = \pi \int_b^a (f(x))^2 dx - \pi \int_b^a (g(x))^2 dx$$

- Disk method: For volume rotated about y axis, consider cylinder with circumference $2\pi x$, height $f(x)$. Volume is the product of circumference and the riemann sum of $f(x)$.

$$V = 2\pi \int_b^a x|f(x) - g(x)| \, dx$$

- Equation of rotation:** The surface obtained rotating f about an axis is $y^2 + z^2 = f(x)^2$ (x-axis); $x^2 + z^2 = f(y)^2$ (y-axis)

Trigonometric substitution

$\int \sqrt{a^2 - (x+b)^2} dx$ (sub with $(x+b) = a \sin \theta$)

$\int \sqrt{a^2 + (x+b)^2} dx$ (sub with $(x+b) = a \tan \theta$)

$\int \sqrt{(x+b)^2 - a^2} dx$ (sub with $(x+b) = a \sec \theta$)

Riemann sums

$$\int_b^a f(x)dx = \lim_{n \rightarrow \infty} \left\{ \sum_{k=1}^n \frac{b-a}{n} f\left(a + k\left(\frac{b-a}{n}\right)\right) \right\}$$

Where $\frac{b-a}{n}$ is the small dx of the function and $f\left(a + k\left(\frac{b-a}{n}\right)\right)$ is the height of the small increment.

Arc length

- Of a Cartesian function:** $S = \int_a^b \sqrt{1 + (f'(x))^2} dx$
- Of a vector valued function:** $L = \int_a^b |r'(t)| \, dt = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} dt$.

Sequences and series

- nth term for divergence:** If $\lim_{n \rightarrow \infty} a_n \neq 0 \vee$ undefined, then series $\sum_{n=1}^{\infty} a_n$ diverges. Inconclusive if $a_n \rightarrow 0$.

- Integral test:** Suppose $a_n = f(n)$; $f'(n) < 0 \forall n > 1$ (decreasing function). If $\int_1^{\infty} f(n)$ converges, so does the series $\sum_1^{\infty} a_n$, and vice-versa. (Reasoning: think of right bounded riemann sum to approximate integral)

- P-series:** The p-series $\sum_1^{\infty} \frac{1}{n^p}$ converges only if $p > 1$.

- Comparison test:** Suppose $0 \leq a_n \leq b_n$. Then,

- $\sum_1^{\infty} b_n$ is convergent $\Rightarrow \sum_1^{\infty} a_n$ converges
- $\sum_1^{\infty} a_n$ is divergent $\Rightarrow \sum_1^{\infty} b_n$ diverges

- Ratio test:** Suppose $\sum_1^{\infty} a_n$, where $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$. If,

- $0 \leq L < 1$, series converge
- $L > 1$, series diverge
- $L = 1$, test inconclusive

- Root test:** Suppose $\sum_1^{\infty} a_n$, where $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L$. If,

- $0 \leq L < 1$, series converge
- $L > 1$, series diverge
- $L = 1$, test inconclusive

- Alternating series test:** If b_n is decreasing and $\lim_{n \rightarrow \infty} b_n = 0$, then alternating series $\sum_1^{\infty} (-1)^n b_n$ converges.

- Absolute convergence test:** If $\sum_{n=1}^{\infty} |a_n|$ is convergent, $\sum_{n=1}^{\infty} a_n$ is also convergent.

Centre/radius of convergence

For a given series $\sum_{n=1}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \dots$, **exactly 1** of the following holds:

- Converge at $x = a$
- Converge for all x
- There exists a positive R where $|x-a| < R$ converges.
 - radius of convergence may take a bounded or unbounded form, eg $(-1, 1]$ or $[-1, 1]$
 - centre of convergence is a .

If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$ (ratio test) or $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L$ (root test), $\exists R = |x+a|$ where $L < 1$.

3d calculus

Vectors

- Component of b along a:** $\text{comp}_a b = |b| \cos \theta = \frac{a \cdot b}{|a|}$
- Projection of b onto a:** $\text{proj}_a b = \text{comp}_a b \times \frac{a}{|a|}$ (multiply by unit vector)
- Dot Product:** $a \cdot b = |a||b| \cos \theta$
 - Commutative and distributive
 - $a \cdot a = |a|^2$
- Cross Product:** $a \times b = (|a||b| \sin \theta) \hat{n}$

- Distributive, $a \times b = -b \times a$

Multivariable functions

- **Partial derivative definition:** $f_x(x,y) = \lim_{h \rightarrow \infty} \frac{f(x+h,y)-f(x,y)}{h}$
- **Clairaut's theorem:** If $f_{xy}(x,y)$ and $f_{yx}(x,y)$ are continuous on D , $f_{xy}(x,y) = f_{yx}(x,y)$.
- **Second derivative test:**

$$D = f_{xx}f_{yy} - \left(f_{xy}\right)^2$$

1. If $D > 0$ and $f_{xx}(a,b) > 0$, then $f(a,b)$ is a local minimum.
2. If $D > 0$ and $f_{xx}(a,b) < 0$, then $f(a,b)$ is a local maximum.
3. If $D < 0$, then (a,b) is a saddle point of f .
4. If $D = 0$, then no conclusion can be drawn.

Properties of functions

Topic	Level Surface $F(x,y,z) = k$	Explicitly Defined Surface $z = f(x,y)$
Finding Tangent Plane Consider 2 vectors on the plane $\begin{pmatrix} 1 \\ 0 \\ f_x(a,b) \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ f_y(a,b) \end{pmatrix} = \begin{pmatrix} f_x(a,b) \\ f_y(a,b) \\ -1 \end{pmatrix}$	$F_x(x-x_0) + F_y(y-y_0) + F_z(z-z_0) = 0$ Vector form: $\nabla F(x_0) \cdot (x-x_0) = 0$	$z-z_0 = f_x(x-x_0) + f_y(y-y_0)$ Vector form: $z-z_0 = \nabla f(x_0,y_0) \cdot (x-x_0,y-y_0)$
Finding Critical Points	$\begin{aligned} F_x(x,y,z) &= 0 \\ F_y(x,y,z) &= 0 \\ F_z(x,y,z) &= 0 \\ F(x,y,z) &= k \end{aligned}$ Vector form: $\nabla F(x) = \mathbf{0} \text{ subject to } F(x) = k$	$\begin{aligned} f_x(x,y) &= 0 \\ f_y(x,y) &= 0 \end{aligned}$ Vector form: $\nabla f(x,y) = \mathbf{0}$

Differentiation

- **Chain rule:** For $u(x_1,x_2,...)$ and $x_1 = f(t_i,...), x_2 = g(t_i,...)$

$$\frac{\partial u}{\partial t_i} = \sum_{j=1}^n \frac{\partial u}{\partial x_j} \frac{\partial x_j}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \cdots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i}$$

- **Implicit Differentiation:** If $F(x,y,z) = k$ defines z implicitly as a differentiable function of x and y , then:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

provided $F_z \neq 0$.

- **Gradient Vector:** Always normal to the surface.

$$\nabla f = \langle f_x, f_y, f_z \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

- **Directional derivative:**

$$D_{\hat{u}}f(x_0,y_0,z_0) = \lim_{h \rightarrow 0} \frac{f(x_0+ha,y_0+hb,z_0+hc) - f(x_0,y_0,z_0)}{h} = \nabla f(x_0,y_0,z_0) \cdot \hat{u} = |\nabla f| \cos \theta$$

- **Differential approximation:** Suppose f is differentiable at (a,b) . Let Δx and Δy be small increments in x and y respectively from (a,b) . Then:

$$\Delta z \approx dz = f_x(a,b)\Delta x + f_y(a,b)\Delta y = f_x(a,b)dx + f_y(a,b)dy$$

provided Δx and Δy are small and $f(x,y)$ is differentiable. Note that the differential dz is the change in the **tangent plane** as (x,y) changes from (a,b) to $(a+\Delta x,b+\Delta h)$.

Double Integrals

- **Definition:**

$$\int \int_R f(x,y) \, \mathrm{d}A = \lim_{m,n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*,y_{ij}^*) \Delta A$$

- **Fubini's Theorem:** If f is continuous on the rectangle $R = \{(x,y) \mid a \leq x \leq b, c \leq y \leq d\}$, then

$$\int \int_R f(x,y) \, \mathrm{d}A = \int_a^b \int_c^d f(x,y) \, \mathrm{d}y \, \mathrm{d}x = \int_c^d \int_a^b f(x,y) \, \mathrm{d}x \, \mathrm{d}y$$

- **Splitting:** If $f(x,y) = g(x)h(y)$,

$$\int \int_R g(x)h(y) \, \mathrm{d}A = \left(\int_a^b g(x) \, \mathrm{d}x \right) \left(\int_c^d h(y) \, \mathrm{d}y \right)$$

where $R = [a,b] \times [c,d]$

- **Iterated Integrals:**

If $D = \{(x,y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$,

$$\int \int_D f(x,y) \, \mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y) \, \mathrm{d}y \, \mathrm{d}x.$$

- **Polar coordinates:**

If f is continuous on a polar rectangle R given by

$$R = \{(r,\theta) \mid 0 \leq a \leq r \leq b, \alpha \leq \theta \leq \beta\}$$

where $0 \leq \beta - \alpha \leq 2\pi$, then

$$\int \int_R f(x,y) \, \mathrm{d}A = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r \, \mathrm{d}r \, \mathrm{d}\theta.$$

- **Jacobian:**

$$\int \int_R f(x,y) \, \mathrm{d}x \, \mathrm{d}y = \int \int_S f(x(u,v),y(u,v)) \left| \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right| \mathrm{d}u \, \mathrm{d}v$$

- **Surface area:** For implicit $F(x,y,z) = k$, make z the subject.

$$\text{Surface Area} = \int \int_D dS = \int \int_D \sqrt{f_x^2 + f_y^2 + 1} \mathrm{d}A$$

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Ordinary Differential Equations

- **Separable DE:** $y' = f(x)g(y)$.

1. $y' = f(\frac{y}{x})$: use $u = \frac{y}{x} \Rightarrow y' = (ux)' = u + u'x$
2. $y' = f(ax+by)$: use $u = ax+by$.

- **Linear first order ODE** $\frac{dy}{dx} + P(x)y = Q(x)$, define $I(x) = e^{\int P(x)dx}$.

$$y \cdot I(x) = \int Q(x) \cdot I(x) dx$$

- **Bernoulli equation:** $y' + P(x)y = Q(x)y^n$, substitute $u = y^{1-n}$. Then get $u' + (1-n)P(x)u = (1-n)Q(x)$, which is first order ODE.